

MATERIAL SAFETY DATA SHEET (MSDS)

Standard GHS Compliance Document for Laboratory Safety

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Chemical Name	Sodium Hydroxide (Caustic Soda)
Chemical Formula	NaOH
CAS Number	1310-73-2
Document Purpose	Academic RAG Validation Dataset

SECTION 2: HAZARDS IDENTIFICATION

Strong base. Causes severe skin burns and serious eye damage. Corrosive to metals.

SECTION 4: FIRST AID MEASURES

- Inhalation:** Remove to fresh air. If symptoms persist, call a physician.
- Skin Contact:** Wash off immediately with plenty of water for at least 15 minutes. Remove contaminated clothing. Seek immediate medical assistance.
- Eye Contact:** Hold eyelids apart and flush eyes with plenty of water for at least 15 minutes. Immediate ophthalmologist examination is mandatory.
- Ingestion:** Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician immediately.

SECTION 6: ACCIDENTAL RELEASE MEASURES

- Precautions:** Avoid dust formation. Avoid breathing vapors, mist, or gas. Ensure adequate ventilation. Wear respiratory protection.
- Clean-up Methods:** Pick up and arrange disposal without creating dust. Sweep up and shovel. Keep in suitable, closed containers for disposal. Carefully neutralize residues with a very dilute acid (like acetic acid).

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Property	Value
Appearance	White solid pellets or flakes
Odor	Odorless
pH	~14 (5% aqueous solution)
Boiling Point	1,388 °C (2,530 °F)
Melting Point	318 °C (604 °F)
Density	2.13 g/cm³

SECTION 10: STABILITY AND REACTIVITY

Strongly hygroscopic. Absorbs moisture and carbon dioxide from the air. Generates heat when dissolved in water. Incompatible with strong acids, aluminum, zinc, and tin (produces flammable hydrogen gas).

SECTION 16: OTHER INFORMATION

This document is synthetically generated for testing Retrieval-Augmented Generation (RAG) system pipelines within clinical biochemistry and safety management training workflows.